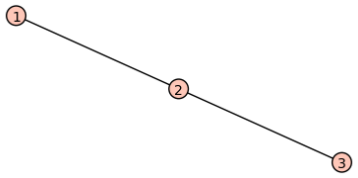
Last name _____

First name _____

LARSON—MATH 756—HOMEWORK WORKSHEET 08
Knuth's *Sandwich Theorem* paper



Real Spectral Theorem Review

1. Find an *adjacency matrix* A for P_3.
2. Find all the eigenvalues, and corresponding normalized eigenvectors for P_3.
3. Find an orthogonal matrix Q and diagonal matrix D so that $A = QDQ^T$. Check all the details.

Characteristic labelings, STAB and QSTAB

For Knuth a *stable set* is a synonym for *independent set*. For a graph G and vertex set $V = \{v_1, \ldots, v_n\}$ ($n = |V|$), a *characteristic labeling* of a set $S \subseteq V$ is a vector $\hat{x}$ in $\mathcal{R}^n$ with $x_v = 1$ if $x_v \in S$ and $x_v = 0$ otherwise.

4. List all of the stable sets in P_3, and find their corresponding characteristic labelings (these are *stable labelings*).

STAB(G) is defined to be the *convex hull* of the stable labels in G. So, if $\hat{x}$ and $\hat{y}$ are in STAB(G) then $t\hat{x} + (1-t)\hat{y}$ is in STAB(G) for every $t \in [0, 1]$ (intuitively this means: for every pair of vectors in STAB(G) the points on the line between them are also in STAB(G).

5. Find examples of non-integer coordinate vectors in STAB(P_3).
6. List all of the cliques in P_3, and find their corresponding characteristic labelings (these are *clique labelings*).

QSTAB(G) is defined to be $\{\vec{x} \geq 0 | \sum_{v \in Q} x_v \leq 1, \text{ for every clique } Q \text{ of } G\} \subseteq \mathcal{R}^n$.

7. Check that all the clique labelings for P_3 are in QSTAB(P_3).
8. Find examples of non-integer coordinate vectors in QSTAB(P_3).
9. Show: for any graph G, STAB(G) $\subseteq$ QSTAB(G).
10. Show: for any graph G, QSTAB(G) is a convex set in $\mathcal{R}^n$.